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PROVISIONAL SPECIFICATION
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AI-DRIVEN SMART INDOOR AIR QUALITY MONITORING AND AUTOMATIC VENTILATION SYSTEM

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The following specification particularly describes the invention and the manner in which it is to be performed.

AI-DRIVEN SMART INDOOR AIR QUALITY MONITORING AND AUTOMATIC VENTILATION SYSTEM

FIELD OF THE INVENTION

The present invention relates generally to the fields of smart environmental monitoring, indoor air quality management, artificial intelligence, Internet of Things (IoT), and intelligent building automation. More particularly, the invention relates to an AI-driven indoor environmental optimization system capable of monitoring, predicting, and controlling indoor air quality conditions in real time through occupancy-aware adaptive ventilation and energy-efficient automation.

BACKGROUND OF THE INVENTION

The present invention relates generally to indoor environmental monitoring systems and, more particularly, to an AI-Driven Smart Indoor Air Quality Monitoring and Automatic Ventilation System capable of continuously monitoring indoor air conditions and automatically regulating ventilation mechanisms using artificial intelligence, IoT communication, occupancy awareness, and predictive analytics.

Indoor environments such as classrooms, conference halls, laboratories, office spaces, libraries, training centres, hospitals, and commercial buildings often experience deterioration in air quality due to high occupancy density, inadequate ventilation, and continuous human activity. Elevated concentrations of carbon dioxide (CO₂), particulate matter (PM_{2.5} and PM₁₀), volatile organic compounds (VOCs), smoke, temperature imbalance, and humidity fluctuations negatively affect human comfort, concentration, cognitive performance, productivity, and overall health. Prolonged exposure to poor indoor air quality may result in respiratory disorders, fatigue, allergies, headaches, cardiovascular complications, and other health-related concerns. Furthermore, many existing ventilation systems operate continuously or at fixed speeds regardless of actual indoor conditions, resulting in excessive energy consumption, inefficient airflow management, and increased operational costs. Conventional solutions also lack adaptive intelligence to differentiate between natural ventilation and mechanical ventilation based on outdoor environmental conditions and occupancy patterns.

Traditional indoor air quality monitoring methods mainly rely on standalone monitoring devices such as carbon dioxide meters, thermometers, humidity sensors, and particulate matter detectors. These devices are generally used independently to measure specific environmental parameters within indoor spaces. Most conventional systems require manual monitoring and human intervention for observing environmental conditions and operating ventilation systems. Existing classroom air monitoring systems generally focus on measuring basic environmental parameters such as temperature, humidity, and carbon dioxide levels using simple sensor-based devices. Some institutions utilize standalone air quality sensors or basic building ventilation systems to improve classroom conditions. However, such systems are typically limited to real-time monitoring only and do not provide predictive analysis, intelligent automation, occupancy-aware environmental control, or adaptive ventilation optimization.

Several earlier inventions have proposed Internet of Things (IoT)-based environmental monitoring systems for indoor spaces. These inventions typically integrate sensors with microcontrollers or embedded systems to measure environmental parameters such as air pollution, temperature, humidity, and gas concentration levels. Many existing inventions utilize wireless communication technologies to transfer environmental data to cloud servers or monitoring dashboards for remote access. Some systems also provide alert mechanisms that notify users when pollutant levels exceed predefined safety thresholds. Certain advanced environmental monitoring systems have incorporated cloud computing, automation technologies, and smart ventilation control mechanisms to improve indoor environmental management. These systems can analyze environmental data, generate reports, and remotely manage ventilation devices through smart building infrastructures.

Several prior patents are related to intelligent indoor air quality monitoring and automated ventilation systems. For example, US Patent No. US12066192B2, titled “Smart Indoor Air Venting System,” discloses an automated ventilation system utilizing environmental sensors to detect indoor air quality events and activate ventilation fans accordingly. Similarly, International Patent WO2020112981A1 describes a smart indoor air venting mechanism configured to monitor pollutant levels and regulate airflow based on predefined environmental conditions. Although these systems provide automated ventilation functionality, they primarily rely on rule-based operations and fixed-threshold activation techniques, thereby limiting adaptive and predictive environmental management capabilities.

Another related invention, US Patent No. US11079130B2, titled “Weather Responsive Smart Ventilation System Using Multiple Optimization Parameters,” introduces ventilation optimization based on weather conditions, humidity, and temperature variations. While the system improves energy efficiency, it lacks advanced AI-driven pollutant forecasting and occupancy-aware adaptive control mechanisms. Likewise, US Patent No. US11156978B2 discloses intelligent HVAC operation control using environmental sensing and operational optimization techniques for building management systems. However, the system mainly focuses on HVAC efficiency and does not comprehensively integrate AI-based indoor air quality prediction with automated smart ventilation control.

Further developments are disclosed in US Patent No. US12018853, which describes AI-assisted air quality management systems utilizing indoor and outdoor environmental datasets for optimizing air exchange strategies. Similarly, Chinese Patent CN114087713B discloses an indoor air quality automatic adjustment system capable of adaptive environmental regulation using localized sensor networks. Some modern inventions also utilize Artificial Intelligence (AI) and Machine Learning (ML) techniques for environmental analysis and anomaly detection. However, such systems are frequently expensive, computationally complex, and primarily intended for commercial buildings, industrial environments, or smart city applications rather than cost-effective classroom and institutional deployments.

Accordingly, there exists a need for an intelligent indoor optimization system capable of predicting future air quality conditions, dynamically adjusting ventilation based on occupancy and environmental context, optimizing energy consumption, selecting the most efficient ventilation strategy, and providing automated and adaptive environmental control. The proposed invention, titled “AI-Driven Smart Indoor Air Quality Monitoring and Automatic Ventilation System,” addresses the aforementioned limitations by integrating real-time environmental monitoring, occupancy detection, cloud connectivity, Artificial Intelligence-based prediction, and automated ventilation control into a single intelligent platform specifically designed for indoor environments such as classrooms, offices, laboratories, and institutional infrastructures.

The system incorporates multiple environmental sensing modules including carbon dioxide (CO₂) sensors, PM2.5 dust sensors, temperature sensors, humidity sensors, and outdoor air quality sensors for continuously monitoring indoor environmental conditions. The invention further integrates occupancy-aware sensing mechanisms, AI-based predictive analytics, cloud-enabled monitoring, intelligent ventilation optimization, and adaptive decision-making capabilities for maintaining healthy indoor air quality while minimizing unnecessary energy consumption. The present invention therefore provides a scalable, energy-efficient, and intelligent indoor environmental management solution suitable for modern smart infrastructure applications.

SUMMARY OF INVENTION

The present invention relates to a smart classroom environmental optimization system (100) for continuously monitoring indoor environmental conditions, analysing occupancy behaviour, predicting future air quality deterioration, and automatically regulating ventilation mechanisms using artificial intelligence and IoT communication technologies. The invention comprises a sensing module (101), a controller unit (102), an IoT cloud communication module (103), a data processing and storage unit (104), a user interface and mobile application module (105), an artificial intelligence prediction module (106), an occupancy detection and behavioural modelling module (107), a ventilation control unit (108), and an alert notification system (109).

The sensing module (101) comprises a plurality of environmental sensors including a carbon dioxide sensor (101A), particulate matter sensor (101B), temperature sensor (101C), humidity sensor (101D), volatile organic compound sensor (101E), and outdoor air quality sensor (101F) for continuously measuring indoor and outdoor environmental parameters. The occupancy detection and behavioural modelling module (107) comprises camera-based human detection units (107A), infrared occupancy sensors (107B), ultrasonic motion sensors (107C), bidirectional counting sensors (107D), and behavioural analytics processors (107E) for determining real-time occupancy density, occupant movement patterns, occupancy variation, and classroom utilisation behaviour.

The controller unit (102) receives environmental data and occupancy information from the sensing module (101) and occupancy detection and behavioural modelling module (107). The

collected information is transmitted through the IoT cloud communication module (103) to the data processing and storage unit (104) for maintaining historical environmental records, occupancy behaviour datasets, pollutant concentration trends, and ventilation activity information. The stored environmental datasets are utilized by the artificial intelligence prediction module (106) for analysing environmental variations, occupancy movement behaviour, classroom usage schedules, and pollutant generation trends.

The artificial intelligence prediction module (106) performs future indoor air quality forecasting by predicting carbon dioxide rise, particulate accumulation, humidity fluctuation, airflow requirements, and environmental deterioration associated with occupancy behaviour and human activity intensity. The artificial intelligence prediction module (106) further generates occupancy-aware environmental forecasting by correlating occupancy density variation with predicted pollutant generation rates and future environmental conditions before hazardous air quality situations occur.

Based on predictive environmental analysis generated by the artificial intelligence prediction module (106), the ventilation control unit (108) performs dynamic ventilation decision-making by selecting natural ventilation, mechanical ventilation, hybrid airflow circulation, or air purification operation according to indoor pollutant concentration, occupancy density, occupancy behaviour, outdoor air quality conditions, and future environmental variations. The ventilation control unit (108) comprises ventilation fans (108A), smart exhaust systems (108B), motorized windows (108C), airflow dampers (108D), HVAC interfaces (108E), air purification devices (108F), and variable-speed airflow regulators (108G) for regulating airflow direction, airflow intensity, ventilation duration, and fresh-air circulation.

The user interface and mobile application module (105) provides real-time environmental visualization, occupancy analytics, predictive air quality information, ventilation activity status, occupancy heatmaps, and alert notifications through cloud-connected dashboards and mobile devices. The alert notification system (109) generates warning notifications during hazardous pollutant accumulation, abnormal occupancy density, ventilation failure, or unsafe indoor environmental conditions.

The present invention therefore provides predictive indoor environmental management, occupancy-aware ventilation intelligence, future air quality forecasting, dynamic ventilation decision-making, adaptive airflow regulation, reduced energy usage, improved indoor environmental quality, and enhanced occupant comfort suitable for smart educational infrastructures and intelligent building environments.

DETAILED DESCRIPTION OF THE INVENTION

The present invention relates to an AI-Driven Smart Indoor Air Quality Monitoring and Automatic Ventilation System (100) for continuously monitoring indoor environmental conditions, analysing occupancy behaviour, predicting future air quality variations, and dynamically regulating ventilation mechanisms using artificial intelligence, IoT

communication, and predictive environmental intelligence. The invention is applicable in classrooms, laboratories, hospitals, office buildings, libraries, industrial spaces, training centres, commercial infrastructures, and residential environments where indoor environmental health and energy-aware ventilation management are required.

In FIG. 1, the system (100) comprises a sensing module (101), a controller unit (102), an IoT cloud communication module (103), a data processing and storage unit (104), a user interface and mobile application module (105), an AI prediction module (106), an occupancy detection and behavioural modelling module (107), a ventilation control unit (108), and an alert notification system (109).

The sensing module (101) comprises a plurality of environmental sensing devices for continuously measuring indoor and outdoor environmental conditions. In one embodiment, the sensing module (101) includes a carbon dioxide sensor (101A), a particulate matter sensor (101B), a temperature sensor (101C), a humidity sensor (101D), a volatile organic compound sensor (101E), and an outdoor air quality sensor (101F). The carbon dioxide sensor (101A) measures indoor CO₂ concentration associated with inadequate ventilation and human respiration density. The particulate matter sensor (101B) detects airborne PM2.5 and PM10 pollutants harmful to respiratory health. The temperature sensor (101C) measures indoor thermal conditions, while the humidity sensor (101D) measures relative humidity associated with indoor comfort and microbial growth conditions. The volatile organic compound sensor (101E) detects gaseous contaminants released from furniture, chemicals, laboratory substances, paints, and cleaning agents. The outdoor air quality sensor (101F) monitors outdoor pollutant concentration, environmental temperature, and atmospheric suitability for fresh-air intake operations.

The occupancy detection and behavioural modelling module (107) is operatively connected with the sensing module (101) and controller unit (102). The occupancy detection and behavioural modelling module (107) comprises camera-based human detection units (107A), infrared occupancy sensors (107B), ultrasonic motion sensors (107C), bidirectional counting sensors (107D), and behavioural analytics processors (107E). The module continuously identifies occupant count, occupant movement patterns, classroom utilisation behaviour, occupancy density variation, seating distribution, activity intensity, and temporal occupancy trends within the monitored environment.

The behavioural analytics processor (107E) generates occupancy heatmaps and occupancy sequence data representing human movement and space utilisation patterns across different time intervals. The occupancy information is continuously transmitted to the AI prediction module (106) for analysing occupancy behaviour and forecasting future indoor environmental variations associated with human presence and activity levels.

The controller unit (102) comprises an edge-processing microcontroller or embedded computing device including ESP32, Raspberry Pi, NVIDIA Jetson, ARM-based processors, or

equivalent embedded systems. The controller unit (102) performs sensor interfacing, data acquisition, signal conditioning, sensor calibration, environmental preprocessing, occupancy data aggregation, and local environmental analysis. The controller unit (102) further performs temporary edge-level decision analysis for reducing cloud communication latency and enabling rapid ventilation response during sudden pollutant spikes or occupancy fluctuations.

The IoT cloud communication module (103) establishes wireless communication between the indoor environmental monitoring system and remote cloud servers using Wi-Fi, Zigbee, GSM, Bluetooth, LoRa, NB-IoT, or equivalent communication technologies. The IoT cloud communication module (103) transmits environmental sensor data, occupancy behaviour information, ventilation activity logs, predictive environmental analytics, and alert notifications to cloud platforms and remote user devices for storage and centralized monitoring.

The data processing and storage unit (104) stores historical environmental records, pollutant concentration trends, occupancy density information, ventilation response behaviour, airflow regulation history, and energy consumption records. The stored datasets are utilized for temporal environmental analysis, occupancy behaviour learning, pollutant forecasting, ventilation performance analysis, and continuous AI model training. The data processing and storage unit (104) further generates environmental timelines and occupancy correlation datasets representing the relationship between occupant activity and indoor air quality deterioration.

The AI prediction module (106) is operatively connected with the data processing and storage unit (104), occupancy detection and behavioural modelling module (107), and ventilation control unit (108). The AI prediction module (106) utilizes artificial intelligence algorithms, machine learning models, neural networks, deep learning frameworks, time-series forecasting models, regression models, reinforcement learning models, and behavioural pattern recognition algorithms for forecasting future indoor air quality conditions.

The AI prediction module (106) analyses historical pollutant trends, occupancy density fluctuations, occupancy movement behaviour, classroom usage schedules, environmental comfort patterns, and outdoor air conditions to predict future pollutant accumulation and environmental degradation before hazardous thresholds occur. The module forecasts future carbon dioxide concentration rise, particulate accumulation, thermal imbalance, humidity variation, and airflow requirements associated with predicted occupancy behaviour.

The AI prediction module (106) further generates occupancy-aware environmental forecasting by correlating occupant count variation with expected pollutant generation rates. For example, the module estimates future carbon dioxide rise associated with increasing classroom occupancy during lecture hours or predicts pollutant reduction during low-occupancy intervals. The AI prediction module (106) continuously updates predictive models according to newly acquired environmental data and occupancy behaviour sequences for improving forecasting accuracy over time.

The ventilation control unit (108) receives environmental prediction outputs and dynamic control instructions from the AI prediction module (106). The ventilation control unit (108) comprises ventilation fans (108A), smart exhaust systems (108B), motorized windows (108C), airflow dampers (108D), HVAC interfaces (108E), air purification devices (108F), and variable-speed airflow regulators (108G). The ventilation control unit (108) dynamically regulates airflow direction, ventilation duration, fan speed, window opening percentage, and air purification intensity according to predicted environmental conditions and occupancy behaviour.

The ventilation control unit (108) performs dynamic ventilation decision-making by selecting between natural ventilation, mechanical ventilation, hybrid airflow circulation, or air purification operation according to indoor pollutant concentration, outdoor air quality conditions, occupancy density, and predicted future air quality behaviour. When outdoor air quality conditions are suitable, the system activates motorized windows (108C) for natural airflow circulation. When outdoor pollution exceeds acceptable conditions, the system activates filtered mechanical ventilation and air purification mechanisms. During high occupancy periods, the ventilation control unit (108) increases airflow circulation intensity according to predicted pollutant accumulation rates.

The ventilation control unit (108) further performs occupancy-responsive airflow adaptation by reducing airflow intensity during low occupancy periods and increasing ventilation response during dense occupancy intervals. The system therefore minimizes unnecessary energy consumption while maintaining healthy indoor environmental conditions.

The alert notification system (109) generates warning notifications during hazardous environmental conditions, abnormal pollutant accumulation, sensor malfunction, ventilation failure, excessive occupancy density, or unsafe thermal conditions. The alert notification system (109) provides visual alerts, audio alarms, mobile notifications, SMS alerts, email alerts, and cloud-based emergency notifications to occupants and administrators.

The user interface and mobile application module (105) provides real-time environmental visualization, occupancy analytics, predictive air quality graphs, ventilation activity status, airflow behaviour analysis, occupancy heatmaps, environmental history timelines, and remote ventilation management functions. The user interface module (105) further enables administrators to monitor future air quality predictions, occupancy forecasting results, environmental alerts, ventilation decisions, and energy usage behaviour through mobile devices or cloud-based dashboards.

In operation, the sensing module (101) continuously acquires indoor and outdoor environmental data while the occupancy detection and behavioural modelling module (107) identifies occupant behaviour and occupancy density patterns. The controller unit (102) pre-processes the acquired data and transmits the processed information to the IoT cloud

communication module (103) and data processing and storage unit (104). The AI prediction module (106) analyses environmental variations, occupancy behaviour sequences, pollutant trends, and historical datasets to predict future indoor air quality conditions and occupancy-associated environmental deterioration.

Based on the predicted environmental behaviour, the ventilation control unit (108) dynamically regulates ventilation systems before indoor pollutant concentration reaches hazardous levels. The ventilation control unit (108) continuously modifies airflow intensity and ventilation strategy according to occupancy fluctuations, predicted pollutant accumulation, and outdoor environmental suitability. Simultaneously, the alert notification system (109) and user interface module (105) provide environmental alerts, predictive analytics visualization, and real-time monitoring information to users.

The present invention therefore provides several technical advantages including predictive indoor environmental management, occupancy-aware ventilation intelligence, future air quality forecasting, dynamic ventilation decision-making, occupancy behaviour analysis, adaptive airflow regulation, reduced energy consumption, improved indoor environmental health, enhanced occupant comfort, and intelligent environmental automation suitable for next-generation smart infrastructures.

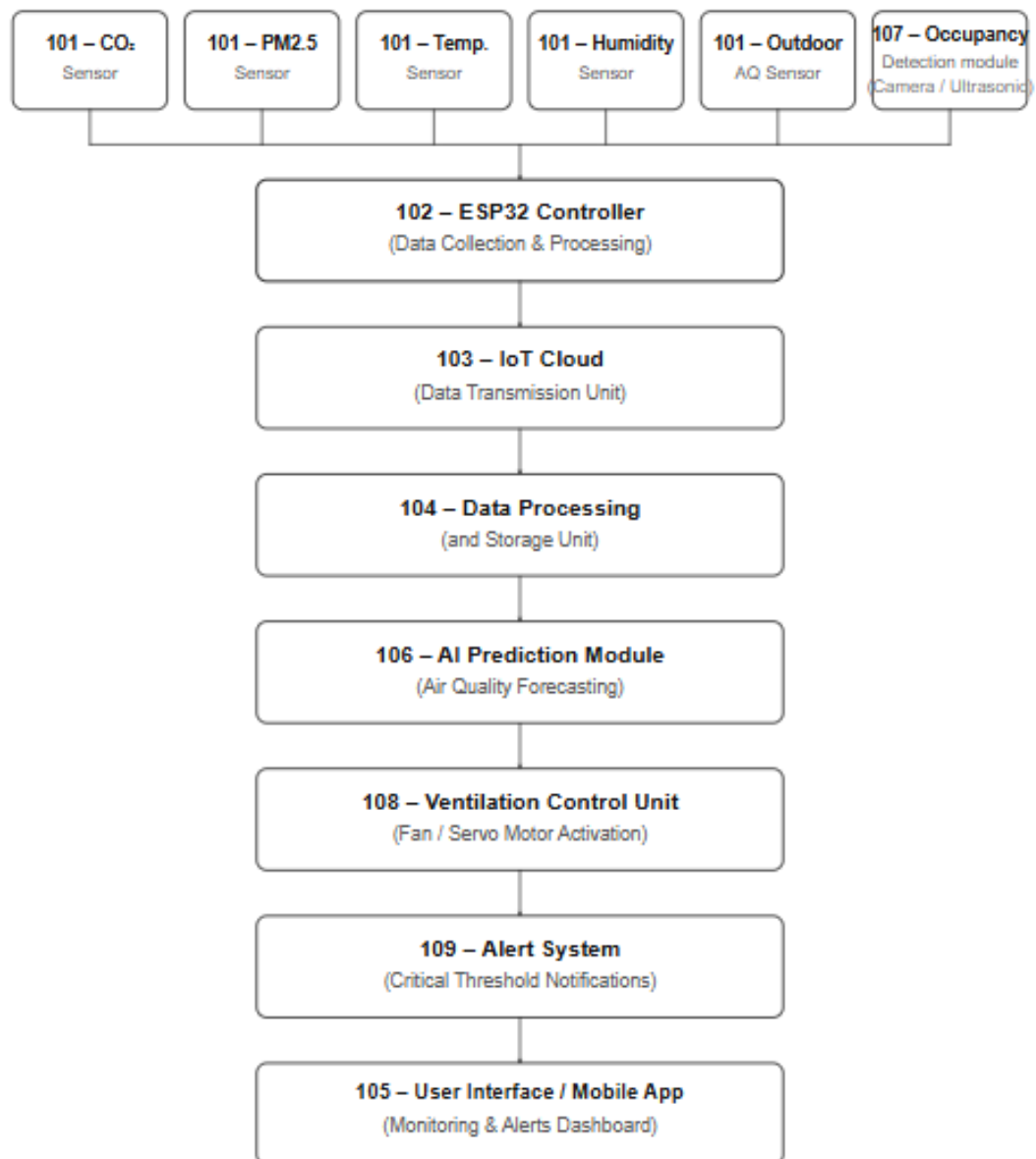
BRIEF DESCRIPTION OF THE DRAWING

FIG. 1 illustrates the overall architecture of the AI-Driven Smart Indoor Air Quality Monitoring and Automatic Ventilation System showing the interconnection between various functional modules used for intelligent indoor environmental monitoring and adaptive ventilation control. The figure depicts the environmental sensing module (101) comprising a carbon dioxide (CO₂) sensor, PM2.5 sensor, temperature sensor, humidity sensor, and outdoor air quality sensor configured for continuously monitoring indoor and outdoor environmental conditions. The figure further illustrates the occupancy detection module (107), which may utilize camera-based sensing, ultrasonic sensing, or motion detection mechanisms for identifying occupancy density and human activity within the monitored environment.

The figure additionally illustrates the ESP32 controller unit (102) configured for sensor interfacing, environmental data acquisition, pre-processing, and real-time signal processing. The controller unit (102) is operatively connected to the IoT cloud communication module (103), which enables wireless transmission of environmental data to cloud-based servers and remote monitoring platforms. The data processing and storage unit (104) shown in the figure is configured for environmental data analysis, historical data storage, and analytics generation. The AI prediction module (106) illustrated in the figure performs machine learning-based air quality forecasting, occupancy-aware environmental analysis, and adaptive decision-making for predictive ventilation optimization.

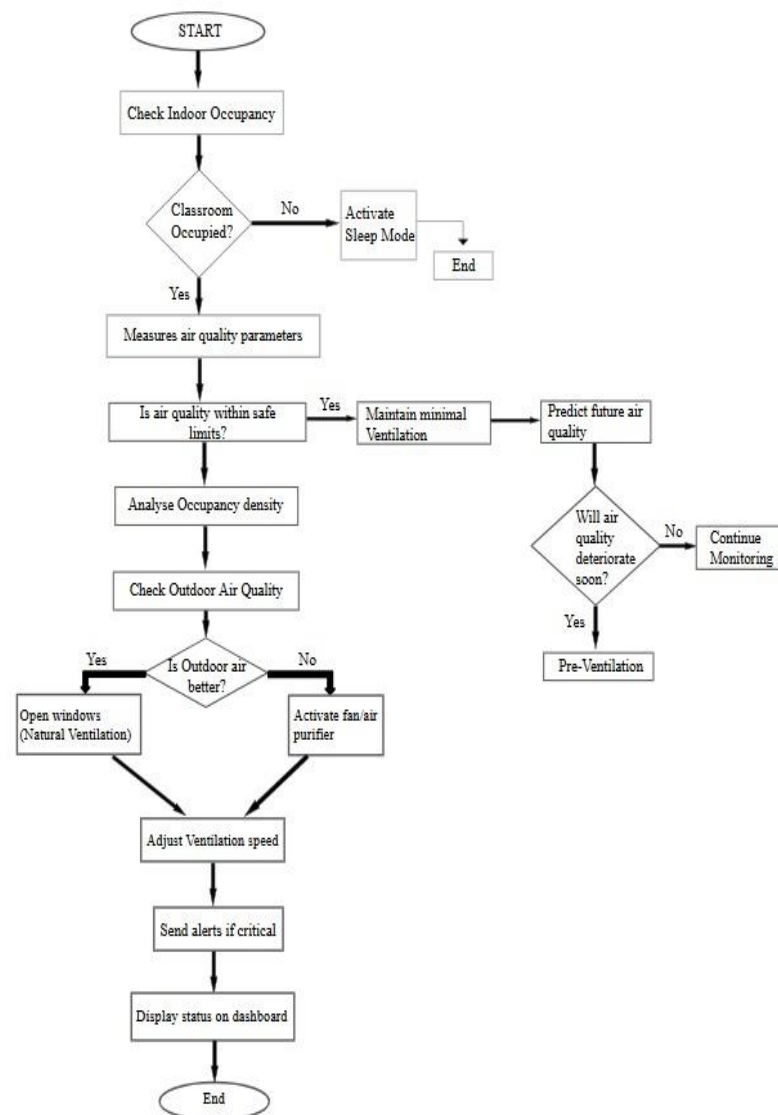
The figure further illustrates the ventilation control unit (108) configured for automatic activation and regulation of ventilation fans, servo motors, HVAC systems, exhaust

mechanisms, and airflow control devices based on AI-generated environmental decisions. The alert notification system (109) illustrated in the figure is configured for generating critical environmental alerts, threshold notifications, and warning indications whenever indoor air quality parameters exceed predefined safety limits. The user interface and mobile application module (105) illustrated in the figure enables real-time indoor air quality monitoring, environmental analytics visualization, remote accessibility, dashboard management, and alert notification display for users and administrators.



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Title: AI-DRIVEN SMART INDOOR AIR QUALITY MONITORING AND AUTOMATIC VENTILATION SYSTEM

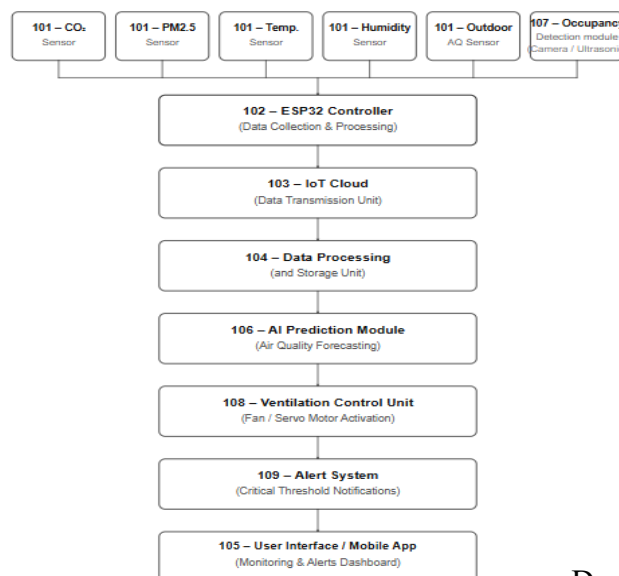
ABSTRACT

The present invention relates to a smart classroom environmental optimization system that integrates Internet of Things (IoT), artificial intelligence (AI), occupancy detection, and automated ventilation control to maintain healthy indoor air quality and improve classroom comfort and energy efficiency. The proposed system continuously monitors environmental parameters including carbon dioxide (CO₂), particulate matter (PM_{2.5}), temperature, humidity, and outdoor air quality using a network of smart sensors. Simultaneously, occupancy detection is performed using computer vision techniques or bidirectional counting sensors to estimate classroom population and occupancy density in real time.

The collected data is processed through an edge computing unit such as ESP32 or Raspberry Pi and transmitted to a cloud-based analytics platform for storage and intelligent processing. An AI-based prediction module utilizing machine learning and time-series forecasting models predicts future air quality conditions based on historical environmental trends and occupancy behaviour. A hybrid classroom optimization engine combines predictive analytics with rule-based decision logic to dynamically determine the most efficient ventilation strategy.

Based on indoor and outdoor environmental conditions, the system automatically selects between natural ventilation through motorized window control and mechanical ventilation using variable-speed fans or air purification systems. The system further incorporates adaptive energy optimization mechanisms, including PWM-based airflow regulation and low-power operation during unoccupied periods, thereby minimizing unnecessary energy consumption. The invention also provides real-time monitoring, alerts, and historical analytics through a web or mobile dashboard interface. By integrating predictive environmental control, occupancy-aware optimization, and energy-efficient ventilation management, the proposed system enhances student comfort, concentration, and indoor safety while reducing manual intervention and operational costs in educational environments.

The invention therefore provides an intelligent, scalable, and automated classroom optimization framework capable of proactive environmental management in smart educational infrastructures.



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We Claim

1. A smart classroom environmental optimization system (100) comprising a sensing module (101), a controller unit (102), an IoT cloud communication module (103), a data processing and storage unit (104), a user interface and mobile application module (105), an artificial intelligence prediction module (106), an occupancy detection and behavioural modelling module (107), a ventilation control unit (108), and an alert notification system (109), wherein the system performs continuous indoor environmental monitoring, occupancy behaviour analysis, future air quality prediction, and automatic ventilation decision-making for maintaining healthy indoor air quality.
2. The system (100) as claimed in claim 1, wherein the sensing module (101) comprises a carbon dioxide sensor (101A), a particulate matter sensor (101B), a temperature sensor (101C), a humidity sensor (101D), a volatile organic compound sensor (101E), and an outdoor air quality sensor (101F) for measuring indoor and outdoor environmental parameters including carbon dioxide concentration, particulate pollutants, thermal conditions, humidity levels, gaseous contaminants, and atmospheric air quality.
3. The system as claimed in claim 1, wherein the occupancy detection and behavioural modelling module (107) comprises camera-based human detection units (107A), infrared occupancy sensors (107B), ultrasonic motion sensors (107C), bidirectional counting sensors (107D), and behavioural analytics processors (107E) for determining real-time occupancy density, occupant movement patterns, occupancy variation, classroom utilisation behaviour, and temporal occupancy trends.
4. The system as claimed in claim 1, wherein the artificial intelligence prediction module (106) analyses historical environmental records, occupancy density information, occupancy movement behaviour, classroom usage schedules, and real-time sensor inputs for predicting future indoor air quality deterioration, pollutant accumulation, carbon dioxide rise, particulate concentration variation, humidity fluctuation, and airflow requirements before hazardous environmental conditions occur.
5. The system as claimed in claim 1, wherein the ventilation control unit (108) performs dynamic ventilation decision-making by selecting natural ventilation, mechanical ventilation, hybrid airflow circulation, or air purification operation according to predicted indoor pollutant concentration, outdoor air quality conditions, occupancy density, occupancy behaviour, and future environmental variations.
6. The system as claimed in claim 1, wherein the ventilation control unit (108) comprises ventilation fans (108A), smart exhaust systems (108B), motorized windows (108C), airflow dampers (108D), HVAC interfaces (108E), air purification devices (108F), and variable-speed airflow regulators (108G) for regulating airflow direction, airflow intensity, ventilation duration, and fresh-air circulation according to predicted environmental conditions and occupancy behaviour while reducing unnecessary energy usage.

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